

C++ Programming

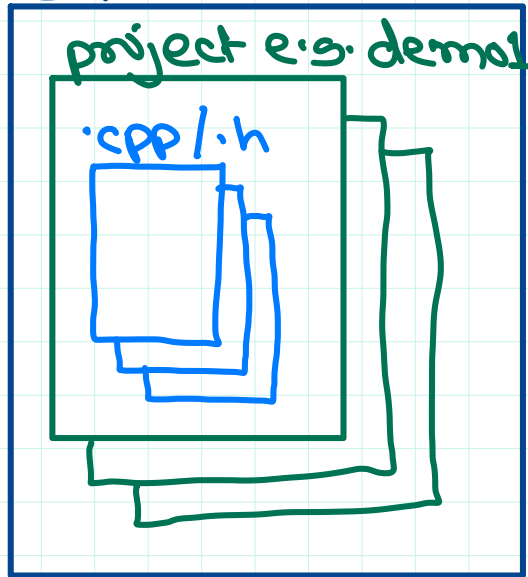
Trainer: Nilesh Ghule

Visual Studio

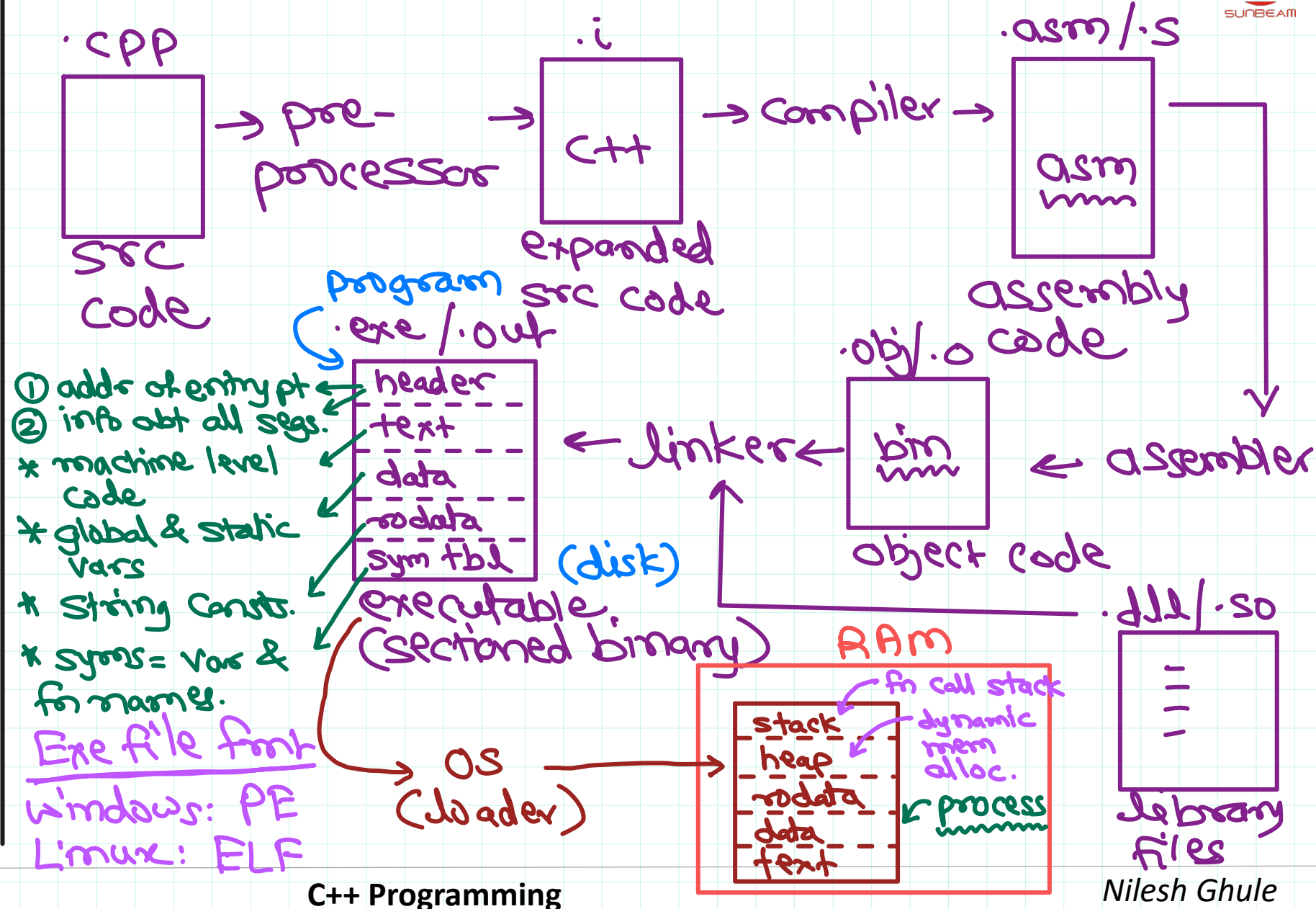
Project & Soln

Soln = group of related projects.

Soln



C++ program Compilation & execution



What is class?

- Building block that binds together data & code
- Program is divided into different classes
- Class has
 - Variables (data members)
 - Functions (member functions or methods)
- By default class members are private(not accessible outside class scope)
- Classes are stand-alone components & can be distributed in form of libraries
- Class is blue-print of an object

What is object?

- Object is an instance of class
- Entity that has physical existence, can store data, send and receive message to communicate with other objects
- Object has
 - Data members (**state** of object)
 - Member function (**behavior** of object)
 - Unique address(**identity** of object)
- The values stored in data members of the object called as 'state' of object
- Behavior is how object acts & reacts, when its state is changed & operations are done
- Behavior is decided by the member functions. Objects behaves differently depending upon the state of the object.
- Operations performed are also known as messages

this pointer

- The invoking object is passed as implicit argument to the function.
- *this* is a keyword in C++.
- *this* pointer points to current object i.e. object invoking the member function.
- Thus every member function receives *this* pointer.

```
class time {  
private: // default - private  
    int hr, min, sec;  
public:  
    void display(){  
        printf("%d:%d:%d",  
            this->hr, this->min, this->sec);  
    }  
};  
  
time t1;  
t1.display();  
time t2(10, 10, 40);  
t2.display();
```

Class data members & member functions

- Data members of the class are generally made as private to provide the data security.
- The private members cannot be accessed outside the class.
- So these members are always accessed by the member functions.
- Member functions are generally declared as public members of class.
- Constructor : Initialize Object
- Destructor : De-initialize Object
- Mutators : Modifies state of the object
- Inspectors : Don't Modify state of object
- Facilitator : Provide additional facilities like IO

Class - constructors

- We can have constructors with
 - No argument : initialize data member to default values
 - One or more arguments : initialize data member to values passed to it
 - Argument of type of object : initialize object by using the values of the data members of the passed object. It is called as copy constructor.
- Constructor is a member function of class having same name as that of class and don't have any return type.
- Constructor get automatically called when object is created i.e. memory is allocated to object.
- If we don't write any constructor, compiler provides a default constructor.

Function overloading

- Functions with same name and different signature are called as overloaded functions.
- Return type is not considered for function overloading.
- Function call is resolved according to types of arguments passed.
- Function overloading is an example of static polymorphism.
- Function overloading is possible due to name mangling done by the C++ compiler

- ```
int sum(int a, int b) {
 return a+b;
}
```
- ```
float sum(float a, float b) {  
    return a+b;  
}
```
- ```
int sum(int a, int b, int c) {
 return a+b+c;
}
```
- Is this overloading??
- ```
float sum(int a, int b) {  
    return a+b;  
}
```




Thank You!

Nilesh Ghule

<nilesh@sunbeaminfo.com>